Symbolic and Distributed Artificial Intelligence

Course Notes

Contents

I Introduction **4**

1 Introduction to Artificial Intelligence **4**

1.1 About the course . . . 4
1.2 What is intelligence? What is AI? . . . 4
1.2.1 The word and a classic definition . . . 5
1.2.2 Many kinds of intelligence . . . 5
1.2.3 Thinking, fast and slow: System 1 and System 2 . . . 6
1.2.4 Artificial intelligence and intelligent agents . . . 7
1.3 A short history of AI . . . 8
1.3.1 Before computers . . . 9
1.3.2 1950: the Turing test . . . 10
1.3.3 1956: the Dartmouth workshop and the founding fathers . . . 11
1.3.4 1966–1972: Shakey the robot . . . 11
1.3.5 1970–1972: Prolog . . . 12
1.3.6 The AI winters and the Lisp machines . . . 12
1.3.7 1997–2016: from chess to Go, the data-driven era . . . 13
1.4 Some more recent achievements . . . 14
1.4.1 Games, medicine and emergent behaviour . . . 14
1.4.2 Self-driving cars: the SAE levels of driving automation . . . 15
1.4.3 Generative AI, protein folding and agentic AI . . . 15
1.5 But... risks, and the answers to them . . . 16
1.5.1 Things that went wrong, or might . . . 16
1.5.2 And hence: principles and guidelines . . . 18
1.6 Symbolic vs. sub-symbolic AI, and what this course is about . . . 18
1.6.1 Two families of AI . . . 18
1.6.2 Thinking fast and slow in AI . . . 19
1.6.3 What SDAI is about . . . 20

II Logic 22

2 Propositional logic 23

3 First-order logic 23

III Agents and Multi-Agent Systems 23

4 Agents and multi-agent systems 23

5 Agent architectures 23

6 Engineering multi-agent systems 23

IV Prolog 23

7 Prolog 23

V Cognitive Agents, Communication and Platforms 23

8 Cognitive agents: BDI and AgentSpeak(L) 23

9 Agent communication 23

10 JADE: a platform for multi-agent systems 23

11 Agent-based modelling and simulation 23

List of whiteboards

The professor wrote these on the whiteboard during the lectures. They are exam material: each one is redrawn in a clean form, and the original photo is shown next to it.

Part I

Introduction

1 Introduction to Artificial Intelligence

[Slides · Thinking machines: 1–97]

Before studying logic, Prolog and agents, the course opens with a big-picture tour: what *intelligence* is, how *artificial* intelligence (AI) was born and grew, what it can do today, why people are worried about it, and where this course sits in the AI landscape. This chapter covers:

1. what intelligence is (several definitions, Kahneman's System 1 and System 2);
2. what artificial intelligence is, and why it is defined through *agents*;
3. a short history of AI, from Aristotle to self-driving cars;
4. recent achievements (AlphaStar, AlphaFold, generative and agentic AI);
5. the risks, and the principles and rules written to control them;
6. symbolic vs. sub-symbolic AI, and why this course is about the *symbolic* and *distributed* side.

1.1 About the course

[Slides · Thinking machines: 1–3]

The course **Symbolic and Distributed Artificial Intelligence** (SDAI, sometimes written S&DAI) is taught by prof. Viviana Mascardi at the University of Genoa. All material (slides, announcements, lab instructions) is published on the course page of the university e-learning platform (AulaWeb).

The course has two souls, already visible in its name:

- **Symbolic**: intelligence obtained by manipulating *symbols* with explicit rules: logic (Part II of these notes) and the logic programming language **Prolog** (Part IV);
- **Distributed**: intelligence spread over many autonomous software **agents** that interact: agents, multi-agent systems, their architectures, their communication and the platforms used to build them (Parts III and V).

Section 1.6 explains, at the end of this chapter, why these two things belong together.

1.2 What is intelligence? What is AI?

[Slides · Thinking machines: 4–12]

To build *artificial* intelligence we should first agree on what intelligence is. There is no single accepted answer; the slides show several complementary views.

1.2.1 The word and a classic definition

[Slides · Thinking machines: 4–6]

The word comes from Latin *intelligĕre*: "to understand", literally "to read inside" (*intus legere*) or "to read between" (*inter legere*), that is, to read beyond the surface of things.

Definition — Intelligence (Mainstream Science on Intelligence, 1994)

Intelligence is a very general mental capability that, among other things, involves the ability to **reason**, **plan**, **solve problems**, **think abstractly**, **comprehend complex ideas**, **learn quickly** and **learn from experience**. It is not merely book-learning, a narrow academic skill, or test-taking smarts. Rather, it reflects a broader and deeper capability for comprehending our surroundings: "catching on", "making sense" of things, or "figuring out" what to do.

The definition was published in the *Wall Street Journal* in December 1994 and signed by 52 scientists. The professor remarks that the statement also contains claims she does not feel comfortable with: it is a reference point, not an undisputed truth. The seven abilities in bold (highlighted in red on slide 6) are worth remembering: each of them has become a branch of AI.

Ability in the definition	AI field that tries to reproduce it
reason	logic, automated reasoning, Prolog (this course)
plan	automated planning (e.g. the STRIPS planner of Shakey)
solve problems	search algorithms, constraint solving
think abstractly	knowledge representation, ontologies
comprehend complex ideas	natural language understanding
learn quickly / from experience	machine learning, reinforcement learning

Table 1: The abilities in the 1994 definition and the AI research areas that study them.

1.2.2 Many kinds of intelligence

[Slides · Thinking machines: 7–8]

Intelligence is not a single number. Two influential theories:

Definition — Multiple intelligences (H. Gardner, *Frames of Mind*, 1983)

Gardner proposed that humans have several distinct intelligences:

1. musical-rhythmic and harmonic;
2. visual-spatial;
3. verbal-linguistic;
4. logical-mathematical;
5. bodily-kinesthetic;
6. interpersonal (understanding others);
7. intrapersonal (understanding oneself);
8. naturalistic (recognising and classifying nature);
9. existential (asking deep questions about life).

Definition — Emotional intelligence (D. Goleman, 1995)

Emotional intelligence (EI) is the capability of individuals to recognise their own emotions and those of others, discern between different feelings and label them appropriately, use emotional information to guide thinking and behaviour, and manage and/or adjust emotions to adapt to environments or achieve one's goal(s).

Intuition — Why does an AI course care?

If intelligence has many faces, then "artificial intelligence" can also mean many different things. A chess program shows logical-mathematical intelligence; a robot that walks on rough ground shows bodily-kinesthetic intelligence; a chatbot that adapts its tone to an angry customer imitates a bit of emotional and interpersonal intelligence. A *multi-agent system* (the main topic of this course) needs *interpersonal* (social) intelligence: agents must understand and negotiate with each other.

1.2.3 Thinking, fast and slow: System 1 and System 2

[Slides · Thinking machines: 9–11]

In the book *Thinking, Fast and Slow* (2011), the psychologist and Nobel prize winner Daniel Kahneman describes the mind as two cooperating "systems".

Definition — System 1 and System 2

- **System 1**: fast, automatic, frequent, emotional, stereotypic, unconscious.
- **System 2**: slow, effortful, infrequent, logical, calculating, conscious.

System 1 can… (in order of complexity)	**System 2 is needed to…**
determine that an object is farther than another	prepare for the start of a sprint
localise the source of a sound	direct your attention to the clowns at the circus
complete the phrase "war and …"	look for the woman with grey hair
show disgust at a gruesome image	try to recognise a sound
solve $2 + 2 =$?	sustain a faster-than-normal walking rate
read text on a billboard	judge if a behaviour is appropriate in a social setting
drive a car on an empty road	count the number of A's in a text
think of a good chess move (if you are a chess master)	give someone your phone number
understand simple sentences	park into a tight parking space
	compare the price/quality ratio of two washing machines
	determine the validity of a complex logical reasoning

Table 2: Kahneman's examples (slides 10–11). Notice that the same task can move from System 2 to System 1 with practice: a chess master "sees" a good move.

Worked example — the bat and the ball

[Not in the slides: added material]

A bat and a ball cost 1.10 euro in total. The bat costs 1 euro more than the ball. How much does the ball cost? System 1 shouts "10 cents!". System 2, if you wake it up, checks: if the ball costs x, the bat costs $x + 1$, so

$$\begin{aligned} x + (x + 1) &= 1.10 \\ 2x &= 0.10 \\ x &= 0.05. \end{aligned}$$

The ball costs 5 cents (check: $0.05 + 1.05 = 1.10$). Fast intuition is cheap but can be wrong; slow reasoning is expensive but verifiable. Keep this in mind: it is exactly the difference between *sub-symbolic* and *symbolic* AI (Section 1.6).

1.2.4 Artificial intelligence and intelligent agents

[Slides · Thinking machines: 12]

Definition — Artificial intelligence

Artificial intelligence (AI) is intelligence demonstrated by machines, unlike the natural intelligence displayed by humans and animals. The leading AI textbook, *Artificial Intelligence: A Modern Approach* (S. Russell and P. Norvig, 4th edition),

defines the field as the study of **intelligent agents**: any device that **perceives** its environment and **takes actions** that maximise its chance of successfully achieving its **goals**.

This definition is the bridge to the whole course: AI is not defined by *how* the device works inside (neural network, logic rules, ...) but by its behaviour as an **agent**, a loop of perceiving and acting towards goals.

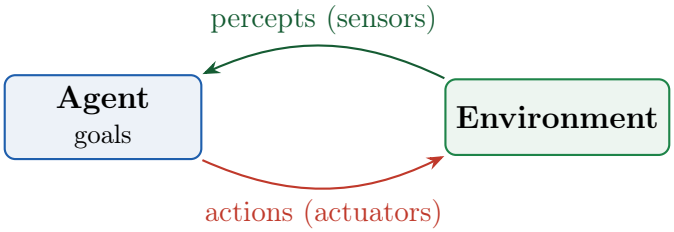


Figure 1: An intelligent agent: it perceives the environment and acts on it to reach its goals. Part III of these notes is entirely about this picture.

Why this matters in practice

Everyday intelligent agents: a robot vacuum cleaner (perceives obstacles and dirt, acts by moving and sucking, goal: a clean floor); a smart thermostat (perceives temperature and your habits, acts on the boiler, goal: comfort at minimum cost); a trading bot (perceives prices, acts by buying/selling, goal: profit); a navigation app (perceives traffic, acts by suggesting a route, goal: get you there fast).

1.3 A short history of AI

[Slides · Thinking machines: 13–67]

The slides build a timeline one event at a time (slides 16–67). The complete timeline is redrawn in Figure 2; the sub-sections below explain each step.

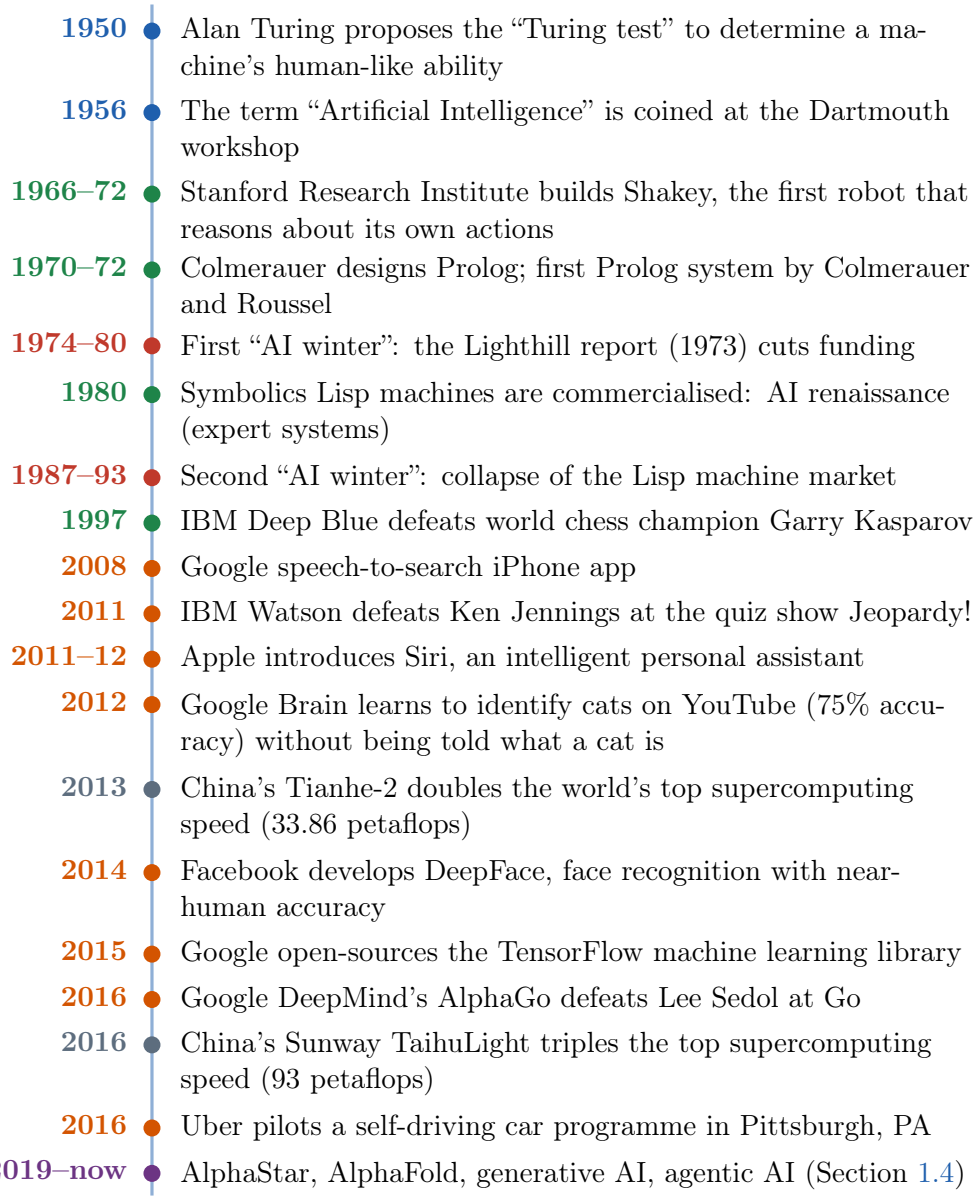


Figure 2: The AI timeline of the slides (slide 67, extended). Colours: foundations, symbolic AI successes, AI winters, data-driven / neural successes, computing power, today. Notice the shift: until the 1990s the big results are symbolic (reasoning, planning, chess search); after 2008 they are mostly learned from data.

1.3.1 Before computers

[Slides · Thinking machines: 13–15]

The slides show three portraits, without text. Here is why each person is on the "before" list (the explanations are added):

- **Aristotle** (384–322 BC) invented the first system of formal logic, the *syllogism*: "All men are mortal; Socrates is a man; therefore Socrates is mortal". The very

same example will come back in first-order logic (Whiteboard **??**) and in Prolog.

- **Gottfried Wilhelm von Leibniz** (1646–1716) dreamed of a universal symbolic language (*characteristica universalis*) and a "calculus of reasoning" (*calculus ratiocinator*), so that disputes could be settled by saying "let us calculate!". He also studied binary arithmetic and built a mechanical calculator. This is the dream of symbolic AI: reasoning as computation.
- **Mary Shelley** (1797–1851) wrote *Frankenstein* (1818), the story of an artificial being created by a scientist, and of the consequences of creating it: the first great story about the risks of artificial creatures (see Section 1.5).

1.3.2 1950: the Turing test

[Slides · Thinking machines: 16–23]

In 1950 Alan Turing published the paper *Computing Machinery and Intelligence* in the philosophy journal *Mind*. Instead of answering the vague question "Can machines think?", he replaced it with a game that can actually be played, the **imitation game**, now called the **Turing test**.

Definition — Turing test

A human interrogator chats (in writing) with two hidden partners: a human and an AI system. By asking questions, the interrogator must decide which one is the machine. If the interrogator cannot reliably tell them apart, the machine passes the test.

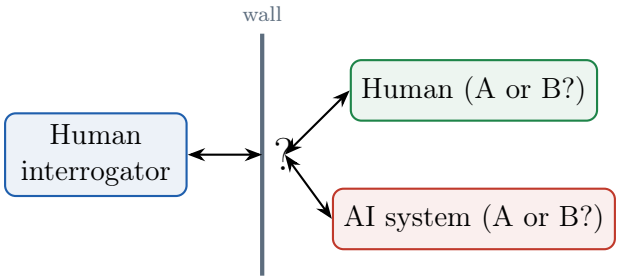


Figure 3: The imitation game (slide 22): the interrogator only sees written answers and must guess which partner is the machine.

Intuition — Behaviour instead of definitions

Turing's trick is to judge intelligence by *observable behaviour*, not by what happens inside. The agent definition of AI (Section 1.2.4) has the same spirit. The slides add a comic ("Turing test 2208") where the human fails: "the machine realised immediately that it was me being the dumb human". Today large language models make the classic Turing test much less decisive than it looked in 1950.

An open parenthesis (slides 19–21). What does it exactly mean to "think", to "perceive", to be "conscious of being alive"? The slides link to the neuroscientist Anil

Seth (TED talk *Your brain hallucinates your conscious reality*): according to him, perception is a *controlled hallucination*, the brain's best guess about the causes of its sensory signals, constantly corrected by the senses. If we cannot define these words precisely for humans, judging them in machines is even harder. (Closed parenthesis.)

1.3.3 1956: the Dartmouth workshop and the founding fathers

[Slides · Thinking machines: 24–33]

In summer 1956 a workshop at Dartmouth College (USA) coined the term **"Artificial Intelligence"**. Its proposal contained the founding conjecture of the field:

> *"Every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it."*

The slides list the participants (the "founding fathers of AI") and, step by step, what each of them is remembered for:

Person	**Known for**
John McCarthy	the *Advice Taker* (a program reasoning with logical sentences, which inspired Prolog), the Lisp language, ALGOL
Marvin Minsky	artificial neural networks, the *society of mind* (the mind as a society of simple interacting agents: an ancestor of multi-agent systems!)
Claude Shannon	information theory, cryptography, natural language processing
Ray Solomonoff	machine learning, prediction, Bayesian frameworks
Allen Newell & Herbert Simon	the *Logic Theory Machine* (proved theorems of *Principia Mathematica*) and the *General Problem Solver*
Arthur Samuel	search, alpha-beta pruning, machine learning (a checkers program that learned to play better than its author)
Oliver Selfridge	Pandemonium theory (pattern recognition by many competing "demons")
Nathaniel Rochester	IBM, designer of the IBM 701 computer
Trenchard More	natural deduction (the proof system of Section **??**)

Table 3: The founding fathers of AI and their contributions (slides 26–33).

Intuition — Symbolic and sub-symbolic, from day one

Already at Dartmouth both souls of AI were present: logic and symbols (McCarthy, Newell and Simon, More) and learning and neural networks (Minsky, Solomonoff, Samuel). The rest of the history is a pendulum between the two.

1.3.4 1966–1972: Shakey the robot

[Slides · Thinking machines: 34–36]

Shakey was the first general-purpose mobile robot able to *reason about its own actions*. Other robots had to be instructed on each individual step; Shakey could analyse a command and break it down into basic chunks by itself. The project combined robotics, computer vision and natural language processing, and was the first to meld *logical reasoning* and *physical action*. It was developed at the AI Center of the Stanford Research Institute (now SRI International) from about 1966 to 1972 (project manager Charles Rosen), programmed mainly in LISP, and its planning component was the **STRIPS** planner. As the first logical, **goal-based agent**, Shakey lived in a limited world (a few rooms with boxes).

Worked example — what "breaking a command into chunks" means

[Not in the slides: added material]

Command: "push the box from room A to room B". A planner like STRIPS knows actions with preconditions and effects, e.g. `goto(Place)`, `push(Box, From, To)`, and finds the sequence: `goto(room_A)` → `goto(box)` → `push(box, room_A, door)` → `push(box, door, room_B)`. Your navigation app and a warehouse robot do the same kind of planning today. Shakey is a deliberative agent in the sense of Section 5.

1.3.5 1970–1972: Prolog

[Slides · Thinking machines: 37]

In 1970 Alain Colmerauer designed the **Prolog** language (from the French *programmation en logique*); the first Prolog system was implemented by Colmerauer and Philippe Roussel in 1972 in Marseille. Prolog is the language of Part IV of these notes: a program is a set of logical facts and rules, and running it means proving a query.

1.3.6 The AI winters and the Lisp machines

[Slides · Thinking machines: 38–43]

Early AI promised a lot. When the promises were not kept, funding collapsed: these periods are called **AI winters**.

- **First AI winter (1974–1980).** The **Lighthill report** (*Artificial Intelligence: A General Survey*, Sir James Lighthill, 1973, for the British Science Research Council) gave a very pessimistic evaluation: "in no part of the field have discoveries made so far produced the major impact that was then promised". It formed the basis for the British government's decision to end support for AI research in all but three universities (Edinburgh, Sussex, Essex). Its main technical criticism: AI had failed to address the **combinatorial explosion** that appears when problems are taken from toy worlds to real-world domains.
- **AI renaissance (1980).** Symbolics commercialises **Lisp machines**: computers with hardware support for Lisp operations (data type testing, CDR coding,

incremental garbage collection), running large Lisp programs very efficiently, with operating systems written entirely in Lisp. They powered the boom of *expert systems* (rule-based programs encoding the knowledge of human experts).

- **Second AI winter (1987–1993).** With the microcomputer revolution, cheap desktop PCs soon ran Lisp programs even faster than Lisp machines, without special hardware. The Lisp machine market collapsed: most makers went out of business by the early 1990s, and AI lost credibility again.

Worked example — combinatorial explosion in numbers

[Not in the slides: added material]

Chess has about 35 legal moves per position. Looking d moves ahead means examining about 35^d positions:

$$35^2 = 1\,225, \qquad 35^4 = 1\,500\,625, \qquad 35^6 \approx 1.8 \times 10^9, \qquad 35^{10} \approx 2.8 \times 10^{15}.$$

Each extra level multiplies the work by 35. An algorithm that works on a toy problem can become hopeless on a slightly bigger one: this is what the Lighthill report meant. Deep Blue (1997) beat it on chess with special hardware plus pruning (alpha-beta, Samuel's legacy).

1.3.7 1997–2016: from chess to Go, the data-driven era

[Slides · Thinking machines: 44–67]

- **1997**: IBM **Deep Blue** defeats the world chess champion Garry Kasparov (symbolic AI: search plus hand-written evaluation).
- **2008**: Google releases a **speech-to-search** iPhone app: you speak, the recording is sent to Google's servers, and the results come back in seconds.
- **2011**: IBM **Watson** defeats the champion Ken Jennings at the quiz show *Jeopardy!* (natural language question answering).
- **2011/2012**: Apple introduces **Siri**, an intelligent personal assistant.
- **2012**: after seeing 10 million images from YouTube videos in three days, the 16,000-computer network of **Google Brain** (one billion connections) began to recognise cats, even though it had never been taught what a cat looked like (unsupervised learning, 75% accuracy).
- **2013**: China's Tianhe-2 doubles the world's top supercomputing speed (33.86 petaflops); **2016**: Sunway TaihuLight triples it (93 petaflops). Raw computing power is one of the reasons for the success of deep learning.
- **2014**: Facebook's **DeepFace** identifies human faces in images: a nine-layer neural network with over 120 million connection weights, trained on four million images uploaded by Facebook users, said to be 97% accurate (vs. 85% for the FBI's Next Generation Identification system). It uses a 3D model to virtually rotate faces so that they look at the camera.
- **2015**: Google open-sources **TensorFlow**, a library for numerical computation with data flow graphs, now widely used for machine learning.

- **2016**: Google DeepMind's **AlphaGo** defeats Lee Sedol at Go, a game whose search space is far too large for Deep Blue-style brute force.
- **2016**: Uber pilots a **self-driving car** programme in Pittsburgh, PA.

Watch out — Two kinds of "winning"

Deep Blue and AlphaGo both beat a world champion, but in very different ways. Deep Blue *searched* millions of positions per second with rules written by humans (System 2 style, symbolic). AlphaGo *learned* an intuition for good moves from millions of games with neural networks, combined with a search (System 1 + System 2). Do not file both under "AI wins at games" without noting the difference: it is the theme of Section 1.6.

1.4 Some more recent achievements

[Slides · Thinking machines: 68–79]

After 2016 the pace accelerated. The slides show a selection of results; for each one, here is what it is and why it matters.

1.4.1 Games, medicine and emergent behaviour

[Slides · Thinking machines: 68–71]

- **AlphaStar** (DeepMind, 2019). StarCraft II is a real-time strategy video game: many units, partial information (you do not see the whole map), real-time decisions. AlphaStar won its historic series against professional players (announced on January 24, 2019, after matches recorded in December 2018; e.g. 5–0 against the pro player MaNa).
- **Lung cancer screening** (*Nature Medicine*, May 2019). A three-dimensional deep learning model analyses low-dose chest CT scans end to end and detects lung cancer with performance comparable to, or better than, expert radiologists.
- **Emergent tool use from multi-agent autocurricula** (OpenAI, ICLR 2020). Agents play *hide-and-seek* in a simulated world with movable boxes and ramps. Nobody tells them how to use the objects; just by competing, the teams discover a sequence of strategies: hiders build shelters with boxes, seekers use ramps to jump in, hiders then lock the ramps away, seekers "surf" on boxes, and so on.

Why this matters in practice — A multi-agent system in action

The hide-and-seek experiment is a *multi-agent system*: several autonomous agents, in a shared environment, competing (hiders vs. seekers) and cooperating (inside a team). Complex behaviour *emerges* from simple goals and interaction, with no central controller. Keep this example in mind for Part III (agents and MAS) and for agent-based simulation (Section 11).

1.4.2 Self-driving cars: the SAE levels of driving automation

[Slides · Thinking machines: 72–74]

How "autonomous" is an autonomous car? The SAE J3016 standard defines six levels.

Level	What the human must do	What the features do	Examples
0	You **are driving**, even if features are on; constantly supervise	warnings and momentary assistance	automatic emergency braking, blind-spot warning, lane departure warning
1	same as level 0	steering **or** brake/acceleration support	lane centering **or** adaptive cruise control
2	same as level 0	steering **and** brake/acceleration support	lane centering **and** adaptive cruise control together
3	You are **not driving** while features are engaged, but you must take over when the system requests it	drives the vehicle under limited conditions	traffic jam chauffeur
4	You are not driving; you will never be required to take over	drives under limited conditions, never needs you	local driverless taxi; pedals and steering wheel may not be installed
5	You are not driving	drives everywhere, in all conditions	as level 4, but anywhere

Table 4: SAE levels of driving automation (slide 72). Blue rows: *driver support* features (the human drives). Green rows: *automated driving* features (the system drives).

The slides note (2024 news) that a **Level 3** car can now be bought in the US: the Mercedes-Benz system works only on a small selection of roads, at limited speed and only in some states, and its wide adoption will require regulators to catch up.

Intuition — Levels of autonomy, again later

The question "how much can the system decide by itself, and when must a human step in?" is the question of **autonomy**, the defining property of agents (Section 4). The jump from level 2 to level 3 is exactly the moment in which responsibility moves from the human to the agent.

1.4.3 Generative AI, protein folding and agentic AI

[Slides · Thinking machines: 75–79]

On slide 75 the professor asked ChatGPT itself: "which are the three most impressive achievements in the AI research field in the last three years (short answer)". The answer:

1. **Generative AI**: models such as GPT-4 and DALL-E generate coherent text, images and creative content.
2. **AI in protein folding**: DeepMind's AlphaFold accurately predicts protein structures, solving a 50-year-old problem in biology.
3. **AI alignment and safety research**: progress in interpretability, alignment and safety (e.g. by OpenAI and Anthropic) to control the behaviour of advanced AI.

Definition — Protein folding problem

A protein is a chain of amino acids. The **protein folding problem** is the question of how a protein's amino acid sequence dictates its three-dimensional atomic structure. It first emerged around 1960, with the first atomic-resolution protein structures.

AlphaFold2 (DeepMind) predicts the 3D structure of proteins from their amino acid sequence with near-experimental accuracy; more than 200 million predicted structures have been released. Knowing a protein's shape tells biologists how it works, which speeds up drug discovery and protein design (2023 review; PNAS 2024, "AlphaFold two years on").

Finally, slide 79: **agentic AI**, presented as "the next evolution of artificial intelligence". An agentic AI system does not just answer one question: given a goal, it plans, calls tools (search engines, APIs, other programs), observes the results and acts again, autonomously, over many steps.

Why this matters in practice — Old ideas, new engines

Agentic AI is the 2020s name for an idea this course has studied for decades: **autonomous agents** that perceive, decide and act towards goals, and **multi-agent systems** in which several of them coordinate. The difference is the engine: a large language model instead of (or together with) explicit beliefs, goals and plans. Understanding classic agents (BDI, communication protocols, platforms) is what makes it possible to engineer agentic AI systems that are reliable.

1.5 But... risks, and the answers to them

[Slides · Thinking machines: 80–87]

The slides now switch tone ("But..."): AI systems already take or support decisions that affect people, and things can go wrong.

1.5.1 Things that went wrong, or might

[Slides · Thinking machines: 80–84]

Case	What happened	The underlying risk
State v. Loomis (2016–17)	The Wisconsin Supreme Court allowed judges to use an algorithmic risk-assessment score (predicting the risk of re-offending) in sentencing, but required a written warning about its limits. The defendant could not inspect how the proprietary algorithm worked.	opaque decisions about people's freedom; possible bias; no way to contest the reasoning
"Smart courts" in China (2019)	Courts adopt AI assistants (e.g. Shanghai's "System 206") to check evidence and support judges.	automation of justice; who is accountable?
ChatGPT banned in Italy (2023)	The Italian data protection authority temporarily blocked ChatGPT over privacy concerns (personal data used for training, no age checks).	privacy and data protection
AI leaders at the UN Security Council	The heads of OpenAI (Sam Altman) and Anthropic (Dario Amodei) gave separate briefings on AI safety to the UN Security Council.	global safety and security
Incidents with AI agents (slide 84)	News reports about security incidents involving AI platforms and AI agents that were misused or hijacked.	autonomous agents acting in the real world can be attacked or misbehave

Table 5: The cautionary examples of slides 80–84.

Intuition — Why symbolic AI helps here

In State v. Loomis the core problem is that nobody could *see* why the system gave a certain score. A symbolic system (a set of logical rules, a Prolog program, a BDI

agent with explicit plans) can *explain* its conclusions: "I concluded X because of rule R applied to facts F_1 and F_2". Explainability, verifiability and accountability are among the strongest arguments for the symbolic side of AI.

1.5.2 And hence: principles and guidelines

[Slides · Thinking machines: 85–87]

- **Asilomar AI Principles** (Future of Life Institute, 2017): 23 principles developed at the 2017 Asilomar conference, grouped into research issues, ethics and values (e.g. safety, transparency, responsibility, human control, privacy), and longer-term issues.
- **Ethics Guidelines for Trustworthy AI** (European Commission, High-Level Expert Group on AI, 8 April 2019). Trustworthy AI must be *lawful*, *ethical* and *robust*, and should meet seven key requirements (listed below, not in the slides).
- **Calls to slow down** (New York Times, September 2026): top AI leaders, among them the head of Anthropic, argued that capabilities are advancing so quickly that greater safety controls are needed across the industry.

Key result — The seven requirements for trustworthy AI (EU, 2019)

[Not in the slides: added material]

1. human agency and oversight;
2. technical robustness and safety;
3. privacy and data governance;
4. transparency (including explainability);
5. diversity, non-discrimination and fairness;
6. societal and environmental well-being;
7. accountability.

These guidelines were followed by the EU *AI Act* (in force since 2024), which regulates AI systems according to their level of risk.

1.6 Symbolic vs. sub-symbolic AI, and what this course is about

[Slides · Thinking machines: 88–97]

The deck ends by going "back to the first slides": Kahneman's two systems (Section 1.2.3) give the key to read the whole history of AI.

1.6.1 Two families of AI

[Slides · Thinking machines: 89–90]

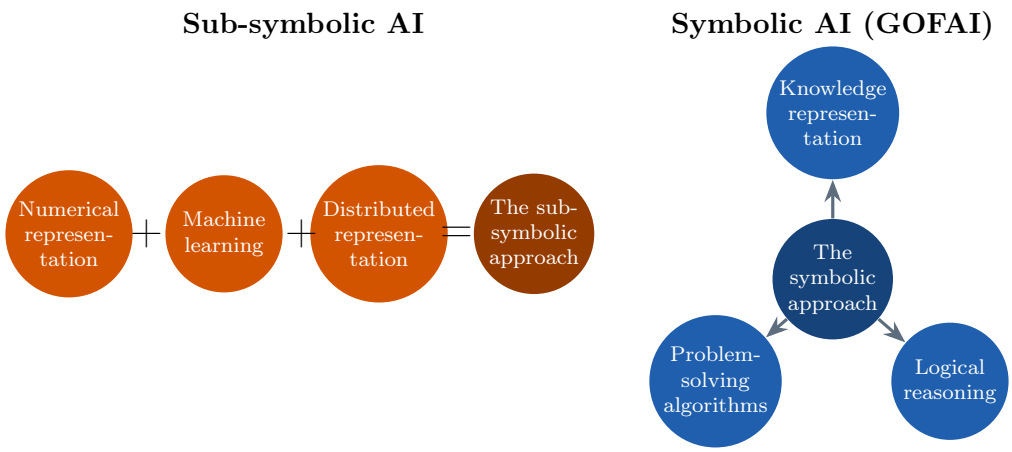


Figure 4: The two families of AI (slides 89–90). Sub-symbolic AI: knowledge is numbers spread over many parameters, learned from data. Symbolic AI: knowledge is explicit symbols, manipulated by reasoning and search.

Definition — Sub-symbolic AI

AI where knowledge is encoded as **numbers** (e.g. the weights of a neural network), **learned** from data (machine learning), and **distributed**: no single number means "cat" or "stop sign"; the meaning is spread over millions of parameters.

Definition — Symbolic AI ("Good Old-Fashioned AI", GOFAI)

AI where knowledge is written explicitly as **symbols** that humans can read (facts, rules, logical formulas, plans), and intelligence is obtained by **knowledge representation**, **logical reasoning** and **problem-solving algorithms** (search, planning).

Worked example — the same knowledge, two ways

Knowledge: "Socrates is a human; all humans are mortal".

- *Symbolic*: two lines of Prolog (Part IV), `human(socrates).` and `mortal(X) :- human(X).` Asking `?- mortal(socrates).` gives `true`, and the system can show the proof: rule + fact.
- *Sub-symbolic*: a language model has read millions of sentences about Socrates and mortality; asked "is Socrates mortal?" it answers "yes" because this is the most likely continuation. The "knowledge" is somewhere in billions of weights and cannot be pointed at.

1.6.2 Thinking fast and slow in AI

[Slides · Thinking machines: 88, 91]

The paper *Thinking Fast and Slow in AI* (G. Booch, F. Rossi and others, AAAI 2021) draws a loose parallel between Kahneman's two systems and the two main lines of work in AI:

	System 1 ≈ machine learning	**System 2 ≈ symbolic AI**
Style	data-driven, fast, intuitive	knowledge-driven, slow, deliberate
Good at	perception from sensory data (seeing, reading, hearing); building (possibly imprecise and biased) models from data	complex problems: logic, search, optimisation, planning, with explicit knowledge, symbols and high-level concepts
Weak at	common-sense reasoning, basic notions of causality, explanations	noisy perception, learning from raw data, scaling to huge open-ended domains
Example	recognise a pedestrian in a camera image	decide that the car must yield because traffic rules say so

Table 6: The parallel of slide 91 ("Thinking Fast and Slow in AI"), with examples.

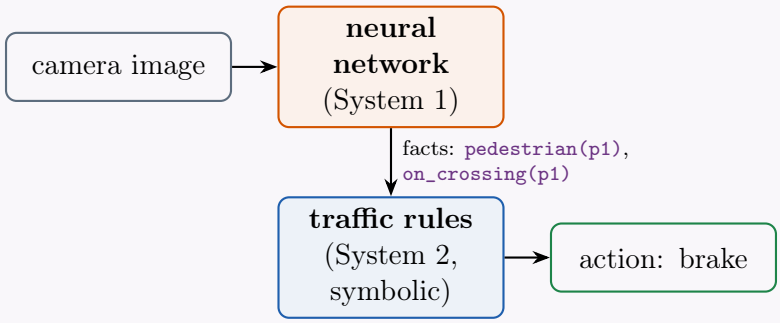
As in humans, the two sets of capabilities are **symbiotic**: this supports **hybrid, neurosymbolic** approaches to AI.

Definition — Neurosymbolic AI

Neurosymbolic AI combines neural networks (sub-symbolic learning) with symbolic representations and reasoning, to obtain systems that are more versatile and more dependable than either alone (slides 92–95: articles in *Communications of the ACM* 2022, IBM Research, the book *Neuro-Symbolic Artificial Intelligence: The State of the Art*, and the journal *Neurosymbolic Artificial Intelligence*, founded in 2023).

Worked example — a neurosymbolic self-driving car

[Not in the slides: added material]



The neural part turns pixels into *symbols* ("there is a pedestrian on the zebra crossing"); the symbolic part applies explicit, auditable rules ("if a pedestrian is on a crossing, then yield"). If the car brakes, we can explain why.

1.6.3 What SDAI is about

[Slides · Thinking machines: 96–97]

Key result — The course in one paragraph (slide 96)

The Symbolic and Distributed AI course is about the **symbolic component of neurosymbolic AI**. It exploits symbolic AI in the context of **distributed AI systems**, where (symbolic, but not only) intelligent software **agents** distributed across different machines operate **autonomously**, to **coordinate/collaborate** in order to achieve some common goal, or to **compete** to access shared resources. A system consisting of these distributed agents, as well as their environment and the organisations they are part of, is named **multi-agent system** (MAS).

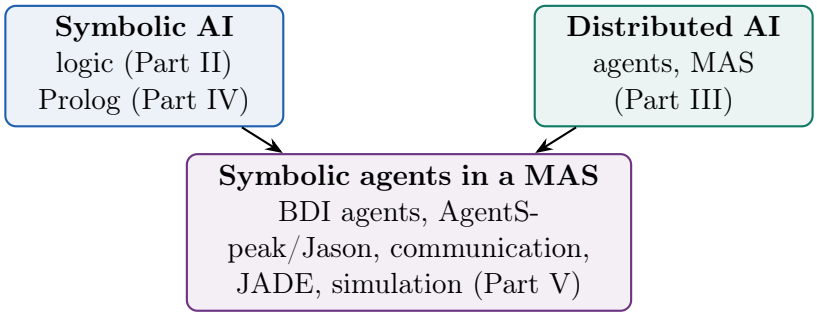


Figure 5: How the parts of these notes fit together.

Welcome on board, and let's have fun with S&DAI!

Cheat sheet — Introduction to AI

- **Intelligence**: general capability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly and learn from experience. Many kinds exist (Gardner's 9 intelligences, emotional intelligence).
- **System 1** = fast, automatic, intuitive; **System 2** = slow, effortful, logical (Kahneman, 2011). Parallel in AI: System 1 ≈ machine learning, System 2 ≈ symbolic AI (logic, planning, search).
- **AI** (Russell & Norvig): the study of *intelligent agents*, devices that perceive their environment and act to maximise the chance of achieving their goals.
- **Key dates**: 1950 Turing test; 1956 Dartmouth workshop (the term "AI"); 1966–72 Shakey; 1970–72 Prolog; 1974–80 and 1987–93 AI winters; 1997 Deep Blue; 2011 Watson; 2012 Google Brain cats; 2016 AlphaGo; 2019 AlphaStar; 2020+ AlphaFold, generative AI; now agentic AI.
- **Risks**: biased decisions (State v. Loomis), privacy (ChatGPT blocked in Italy), safety and security of autonomous agents. **Responses**: Asilomar principles (2017), EU Ethics guidelines for trustworthy AI (2019).
- **Sub-symbolic AI** = numerical representation + machine learning + distributed representation. **Symbolic AI** (GOFAI) = knowledge representation + logical reasoning + problem-solving algorithms. **Neurosymbolic AI** combines them.
- **SDAI** = the symbolic component of neurosymbolic AI, used inside distributed systems of autonomous agents: a **multi-agent system (MAS)**.

Part II

Logic

2 Propositional logic

3 First-order logic

Part III

Agents and Multi-Agent Systems

4 Agents and multi-agent systems

5 Agent architectures

6 Engineering multi-agent systems

Part IV

Prolog

7 Prolog

Part V

Cognitive Agents, Communication and Platforms

8 Cognitive agents: BDI and AgentSpeak(L)

9 Agent communication

10 JADE: a platform for multi-agent systems

11 Agent-based modelling and simulation